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Bullet-screen engagement in videos and deindividualized online behavior: the chain mediating role of belonging and loneliness

Mingxuan Zou[1,2] · Defang Mu[3]

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Abstract
The bullet-screen function is frequently used by individuals when watching online videos. It can enhance the viewers' sense of belonging to the bullet screen group and reduce their feelings of loneliness while watching online videos. However, the deindividuation phenomenon often occurs on the network platform with user interaction functions. As a popular interactive video technology, it is unclear whether the bullet-screen engagement in videos will affect the viewers' deindividualized online behavior. Based on the theory of computer-mediated communication and the social identity model of deindividuation effects, this study explored the relationship between bullet-screen engagement and deindividualized online behavior, and the role of sense of belonging and loneliness between them. A total of 511 college students in China (M = 21.53, SD = 2.02, 45.4% males) were investigated with The Bullet-screen Engagement Questionnaire, The Sense of Belonging Instrument, The Loneliness Scale, and The Bullet-screen Environment Deindividualized Online Behavior Questionnaire. Sequential chain mediation model was built by the macro program PROCESS in SPSS. The model showed that (1) bullet-screen engagement positively predicted deindividualized online behavior; (2) The sense of belonging and loneliness mediated the relationship between bullet-screen engagement and deindividualized online behavior respectively; (3) The sequential chain mediation showed an indirect path (bullet-screen engagement – sense of belonging – loneliness – deindividualized online behavior). Bullet-screen engagement in videos can enhance viewers' sense of belonging and reduce loneliness, by which can also lead to deindividualized online behavior. Therefore, it is also crucial to pay attention to the negative effects of bullet-screen engagement in video.

Keywords Bullet-screen · Deindividualized online behavior · Belonging · Loneliness

Introduction

As a novel video interaction technology, bullet-screen is a commentary subtitle edited by viewers in different spatial and temporal conditions, and it appears on the top of the video screen in real time as scrolling, lingering, or even more action effects while watching a video (Yan et al., 2023; Yang, 2020; Hao et al., 2019). Bullet-screen engagement in videos refers to the degree of viewer involvement with the bullet screen content while watching online videos (Li, 2020). The degree of involvement determines the level of engagement in the bullet screen and the extent of use of its functions. Bilibili site (https://www.bilibili.com/) is a typical bullet screen website in China. Until 2021, the total number of bullet-screens on Bilibili site has reached 10 billion, with more than 60 million bullet-screen users and an average of 166 comments per user (Yawei, 2021).

Bullet screen engagement in the video viewing process has a positive effect. For example, the process of bullet-screen presentation has the characteristics of timeliness and interactivity, and bullet-screens' content is very rich and interesting, which are displayed at the top of the screen in real time, including not only video-related content, but also video-unrelated content (Zhou et al., 2019). Thus, bullet screen can provide viewers with more supplementary

✉ Defang Mu
mudefang2013@163.com

Mingxuan Zou
z1291014528@163.com

1 Public Health School, Shaanxi University of Chinese Medicine, Xi'an 712046, China

2 Department of Military Medical Psychology, Air Force Medical University, Xi'an 710032, China

3 Department of Psychology, Tianjin University of Commerce, Tianjin 300134, China

information not found in video materials and meet their information acquisition needs (Gong & Yan, 2023). At the same time, the real-time display of the bullet screen will create a more interactive virtual network community, which will improve the viewers' sense of immersion and belonging during the video watching process, slow down the individual's subjective feeling of time passing, and reduce the sense of loneliness (Chen et al., 2019a). Therefore, they also can meet the viewers' needs for social enhancement (Zeng et al., 2023).

However, bullet screen engagement in video also has negative effects, such as the occurrence of deindividualized online behavior (Shih & Huang, 2014). The Internet is regarded as a "third space" between physical and mental space, in which people exhibit psychological and behavioral characteristics that are different from those of the real world. Deindividualization (namely depersonalization) is a phenomenon of "Loss" of individuality in a group, which is manifested by a decline in the awareness and control of one's own behavior and the emergence of behaviors that do not occur when one is alone, it is usually done in an abnormal, unruly way (Zimbardo, 1969). Deindividualized online behavior is the disinhibition behavior that occurs when an individual reduces his sense of social responsibility in an online environment, such as making offensive and inflammatory comments in an online environment, the organization network bullying and other negative behavior (Vallury et al., 2021; Spears, 2017). Deindividualized online behavior is very dangerous and pervasive. For example, the microblogging platform in China, which is popular among youth groups, is often characterized by online misconduct such as spreading rumors and insults, and these phenomena can have adverse effects on individuals and society (Yi et al., 2019). However, there is a lack of research on the relationship between bullet-screen engagement and deindividualized online behavior in video. In other words, it is unclear whether the bullet-screen engagement will affect the deindividualized online behavior. Thus, this study explored the relationship between bullet-screen engagement and deindividualized online behavior, and the influence mechanism between them.

The potential relationship between bullet-screen engagement and deindividualized online behavior

Traditional video social media communities are mostly non-synchronous "building" comments (for example, forums, blogs, etc.), lacking immediacy and synchronicity (Gong & Yan, 2023). As mentioned above, with the development of information technology, bullet-screens, a new video interaction technology, can create more interactive virtual online communities for the viewers (Zhou et al., 2019). But at the same time, it also has become a new platform for the occurrence of deindividualized online behaviors (Johnson, 2013; Yang, 2020). Currently, there are no studies directly investigating the relationship between the two. However, relevant theories can provide a basis for understanding their potential relationship.

According to Computer-Mediated Communication (CMC) theory and the social identity model of deindividuation effects (SIDE), when the viewer watching videos, the deeper engagement into the content of the bullet screen, the easier for individuals to identify with the norms of the virtual group, and generate deindividualized online behavior (Lan & Zuo, 2009). Previous studies have shown that there are differences in the degree of engagement into bullet-screen content for each video viewer (He, 2017), such as some viewers just passively watch without actively posting, while others actively post, like the comment, or reply. Different levels of engagement may produce different psychological interactions (Johnson, 2013; Chen et al., 2019a). Watching without participation is a kind of passive psychological interaction, and when individuals send and receive "Instant" feedback from other viewers in space and time, the degree of their engagement to the content of the bullet screen significantly increases. These viewers who have higher bullet-screen engagement in video would be so immersed in the bullet-screen content that will ignore the passing of time and decrease their cognitive resources to the bullet-screen video, which will greatly promote the social identity of the virtual group, and then lead to deindividualized online behavior (Shih & Huang, 2014), especially when the feedback content is highly consistent with the original attitude of the viewers. Besides, the anonymity of the bullet-screen sending process, this feature that also attracts individuals to deeply participate in and commit to the content of the bullet-screen, and prompts them to express their identification with the group, reduces their sense of responsibility for their speech, resulting in deindividualized online behaviors (Lan & Zuo, 2009). Therefore, Hypothesis 1 is proposed: when individuals watch videos, the level of engagement in bullet-screens positively affects their deindividualized online behaviors.

The mediating role of belonging

Acquirement of belonging may have a mediated role on the relationship between bullet-screen engagement and deindividualized online behavior. For one thing, engagement in bullet-screen content can somewhat increase an individual's sense of belonging. The social identity theory of deindividuation suggests that when an individual is involved in a real or virtual group, there is a process of switching from personal to social identity, in which the individual is led to develop belonging emotions that promote the development of health and well-being (Greenaway et al., 2016). Studies have been conducted to investigate the motivation for

using bullet-screens, and the results also found that the main motivation for viewers to use bullet-screens is the desire to gain a sense of self-belonging within the bullet-screen group (Chen et al., 2015). Through the structural equation model, the research of Xiang and Chae (2022) also found that bullet screen can improve the sense of belonging. Sense of belonging refers to the emotional experience that arises from an individual's participation in an environment or group, feeling a special role that he or she plays in this environment or group, and the individual's desire to fit in, which can be influenced by the group environment (Zeng et al., 2023).

For another thing, an increased sense of belonging may promote deindividualized behavior. Studies indicate that the bullet-screen environment allows individuals to communicate freely in a two-way manner, obtaining bullet-screen supplementary information and get immediate feedback, so this virtual environment can enable viewers to develop the sense of belonging and to acquire and identify with the norms of the bullet-screen group by increasing the sense of belonging (Xi et al., 2021), which then have a tendency to display deindividualized online behaviors that are consistent with the norms (Shih & Huang, 2014). Empirical studies have also pointed out that the sense of belonging can affect the deindividualized behavior, satisfying the need of belonging can promote the deindividualized online behavior (Chen et al., 2019b). Therefore, the study proposes Hypothesis 2: Sense of belonging has a mediating role between the level of bullet-screen engagement and deindividualized online behavior.

The mediating role of loneliness

In addition to enhancing the viewers' sense of belonging, bullet screens can also reduce their sense of loneliness during the video viewing process, creating the impression that many people are watching the video with the viewer at the same time (Chen et al., 2019a). Loneliness is a negative affective experience, which occurs when an individual senses a discrepancy between the actual socialization pattern obtained and the expectations (Hawkley & Cacioppo, 2010). According to computer-mediated communication theory, the bullet-screen environment has the characteristic of "Pseudo-synchronization", in which emotional communication can achieve the effect of face-to-face communication, it can even make viewers communicate more frequently and more engaged than they would in a real social environment (Gedvilaitė-Kordušienė & Mikulionienė, 2023). Thus, engagement in bullet-screen content can reduce the loneliness of viewers during the video viewing (Johnson, 2013). Previous studies have also shown that bullet-screen engagement can reduce viewers' loneliness level (He, 2017; Chen et al., 2019a). For example, with the help of eye-tracking technology, a researcher observes the gaze of lonely viewers and ordinary viewers on different appealing information (emotional or rational) under the conditions with or without bullet-screen video, in order to explore the degree of engagement of different viewers to bullet-screen content and their eye-tracking mechanisms, and the results show that bullet-screen can reduce the loneliness level of lonely viewers in online communication and enhance their identification with this virtual group (Chen et al., 2019a).

According to social identity theory, loneliness is related to an individual's lack of an important social group and corresponding social identity (Hayes et al., 2022), bullet-screens provide a social group for an individual and reduce the viewer's loneliness level and elevate the viewer's social identity with the group. Furthermore, according to the social identity model of deindividuation effects, when an individual satisfies the missing social group and social identity needs, they will show the tendency of deindividualized behaviors to follow the rules of the group, which promotes the generation of deindividualized online behaviors (Zeng et al., 2023). Therefore, Hypothesis 3 is proposed: when individuals watch bullet-screen videos, a decrease in loneliness mediates the relationship between bullet-screen engagement and deindividualized online behaviors.

The chain mediating effect of belonging and loneliness

Watching bullet-screen in videos can also create an emotional communication. Emotional communication can cause individuals to have corresponding emotions, which can affect the behavior of individuals, such as deindividualized online behavior (Planalp & Rosenberg, 2014). Belonging and loneliness are both emotions that arise when individuals interact and communicate with real or virtual groups, they are two of the emotions that play antagonistic roles, that is, a high level of belonging is accompanied by low loneliness or a low level of belonging is accompanied by high loneliness (Lim et al., 2021). Empirical studies have shown that there is a potential causal relationship between the individual's sense of belonging, the reduced level of loneliness, and deindividualized behavior, and that the satisfaction of the need to belong can promote an individual's deindividualized online behavior by reducing his or her loneliness (Chen et al., 2019b). It can be seen that belonging and loneliness reduction may have a more complex influence path between the level of bullet-screen engagement and deindividualized online behavior. Therefore, this study proposes Hypothesis 4: The degree of bullet-screen engagement influences deindividualized online behavior through the chain mediation of increased sense of belonging and decreased sense of loneliness. Figure 1 displays the research hypothesis model.

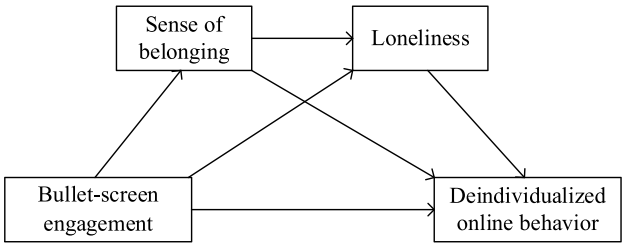

Fig. 1 The research hypothesis model

Methods

Participants

The study employed the convenient sampling method. A total of 522 college students from three universities in Tianjin, Guangzhou and Lanzhou in China were selected for the study. After excluding the subjects who had not used the bullet screen at all, there were 511 valid subjects (age: $M = 21.53$, $SD = 2.02$), and the questionnaire recovery rate was 97.9%. Of these, 232 were male (age: $M = 21.45$, $SD = 2.03$) and 279 were female (age: $M = 21.60$, $SD = 2.02$); 319 were only-born (62.4%) and 192 were not only-born (37.6%); 33(6.5%) used bullet screens for less than 1 year, 177(34.6%) for 1–2 years, 194(38.0%) for 3–4 years, and 107(20.9%) for more than 5 years.

Measures

The bullet-screen engagement questionnaire

The research adopts The Bullet-screen Engagement Questionnaire, which mainly examines the participation and immersion degree of bullet screen for viewers when they watch the video (He, 2017). Likert 5 points are used, where 1 represents "Never" and 5 represents "Always", and the higher the score, the greater the individual's commitment to the bullet screen. The alpha of questionnaire consistency was 0.887.

The sense of belonging instrument

The Sense of Belonging Instrument-Psychological State (The SOBI-P), developed by Hagerty and Patusky (1995), was used to assess an individual's sense of comfort and inclusion in a system or environment, the psychological state of feeling valued, accepted, and needed by the group can be measured. This scale consists of 18 questions, which are scored by Likert 5 points. The total score ranges from 18 to 90. The higher the score, the higher the sense of belonging. The alpha of questionnaire consistency was 0.944.

The loneliness scale

The Loneliness Scale adapted by He (2017) based on the environment of bullet-screen was used to measure the extent to which the bullet-screen reduces the level of individual loneliness. The lower the score is, the higher the degree of the bullet screen engagement eliminates the individual loneliness, the lower the level of loneliness experienced in the bullet screen environment. The questionnaire consisted of 7 items, which were scored by Likert 5 points, 1 for "Very non-conforming", 5 for "Very conforming", and all items were scored in reverse. The alpha of questionnaire consistency was 0.912.

The bullet-screen environment deindividualized online behavior questionnaire

The Depersonalization Dimensions in the questionnaire of influencing factors of network cluster behavior compiled by Chen (2019) was used to measure one's deindividualized online behavior in bullet screen environment. The questionnaire includes 8 questions, using a Likert 5-point scale, with higher total scores representing individuals who are more likely to produce deindividualized online behaviors in the bullet screen environment. The consistency reliability of the questionnaire was 0.903.

Data collection and statistical analysis

All participants completed informed consent and psychological professionals were arranged in the universities of the three regions to organize and distribute questionnaires. During the testing process, the personnel provided guidance for completing the questionnaire. Subsequently, participants filled out questionnaires and returned them to the personnel, and were paid. Data were analyzed using SPSS 23.0 and the macro program PROCESS 3.3 in SPSS, and the significance of regression coefficients was tested using the bias-corrected nonparametric percentile bootstrap method (Hayes, 2017).

Results

Common method biases test

The common method biases test conducted by Harman's one-factor method showed that the exploratory factor analysis extracted five factors (*Eigenvalues* > 1), and the largest factor explained 34.32% of the variance, which complied with the critical condition of less than 40% (Zhou & Long,

2004), indicating that the study did not suffer from serious common method bias.

Descriptive statistics and related analysis

The correlation analysis of the degree of bullet-screen engagement, sense of belonging, reduced level of loneliness and deindividualized online behavior showed that the degree of bullet-screen engagement was significantly positively correlated with deindividualized online behavior and sense of belonging, and significantly negatively correlated with loneliness, and that loneliness was significantly negatively correlated with both sense of belonging and deindividualized online behavior. The results are shown in Table 1.

Mediating effect test

Combining the research hypotheses, used deindividualized online behavior as the outcome variable, the degree of bullet screen engagement as the predictor variable, and sense of belonging and loneliness as the mediator variables, and selected the Model 6 model (chained mediation with two mediator variables) in PROCESS to conduct the chained mediation effect test. The results of regression analysis showed that the degree of bullet screen engagement positively predicted sense of belonging and deindividualized online behavior (sense of belonging: $\beta = 0.16$, $t = 3.58$, $p < 0.001$, 95% CI [0.15, 0.53]; deindividualized online behavior: $\beta = 0.38$, $t = 6.96$, $p < 0.001$, 95% CI [0.27, 0.49]), and negatively predicted loneliness, $\beta = -0.83$, $t = -33.65$, $p < 0.001$, 95% CI [–0.73, – 0.65]. Sense of belonging positively predicted deindividualized online behavior, $\beta = 0.19$, $t = 6.29$, $p < 0.001$, 95% CI [0.06, 0.12], and negatively predicted loneliness, $\beta = -0.06$, $t = -2.38$, $p = 0.018$, 95% CI[–0.04,–0.01]. In addition, loneliness also negatively predicted deindividualized online behavior, $\beta = -0.34$, $t = -6.14$, $p < 0.001$, 95% CI[–0.54,–0.28]. The results are shown in Table 2.

The results of mediation effect analysis showed that sense of belonging and loneliness mediated significantly between the level of bullet screen engagement and deindividualized online behaviors, and that sense of belonging and loneliness acted as chained mediation between the level of bullet screen engagement and deindividualized online behaviors. The results are shown in Table 3. The total indirect effect value was 0.31. which accounted for 44.67% of the total effect value. The chain mediating role is shown in Fig. 2.

Discussion

Based on computer-mediated communication theory and the social identity model of deindividuation effect, this study explored the relationship between the degree of engagement to bullet screens and the deindividualized online behaviors by individuals, as well as the mediating roles of sense of belonging and loneliness between the two. The results of research found that the content of engagement to bullet screen not only directly and positively predicts the occurrence of deindividualized online behaviors when watching bullet screen videos, but also predicts the occurrence of deindividualized online behaviors through the two separate

Table 1 Descriptive statistics and correlation analysis results of variables

	M	SD	1	2	3	4
1. Bullet screen engagement	32.20	7.29	—			
2. Sense of belonging	65.13	15.87	0.16**	—		
3. Loneliness	14.92	6.07	-0.84**	-0.19**	—	
4. Deindividualized online behavior	28.67	7.36	0.69**	0.31**	-0.69**	—

**$p < 0.01$

Table 2 Regression analysis of the relationship between variables in the chain mediation model

	Sense of belonging			Loneliness			Deindividualized online behavior		
Variant	β	*SE*	*t*	β	*SE*	*t*	β	*SE*	*t*
Bullet screen engagement	0.16	0.10	3.58***	-0.83	0.02	–33.65***	0.38	0.05	6.96***
M1: Sense of belonging				-0.06	0.01	-2.38*	0.19	0.01	6.29***
M2: Loneliness							-0.34	0.07	-6.14***
R	0.16			0.84			0.74		
R^2	0.02			0.70			0.55		
F	12.85***			596.18***			206.07***		

β values in the table are standardized coefficients, *$p < 0.05$, ***$p < 0.001$

Table 3 Bootstrap analysis of the mediated effects test

	Efficiency value	Boot standard error	Lower limit of 95% CI	Upper limit of 95% CI
Aggregate effect	0.694	0.033	0.001	0.630
Direct effect	0.382	0.055	0.001	0.274
Total indirect effect	0.310	0.073	0.164	0.453
Path 1: Bullet screen engagement → sense of belonging → deindividualized online behavior	0.030	0.012	0.011	0.057
Path 2: Bullet screen engagement → loneliness → deindividualized online behavior	0.277	0.073	0.132	0.420
Path 3: Bullet screen engagement → sense of belonging → loneliness → deindividualized online behavior	0.003	0.002	0.001	0.008

Total indirect effects and paths 1–3 are fully standardized coefficients in the table

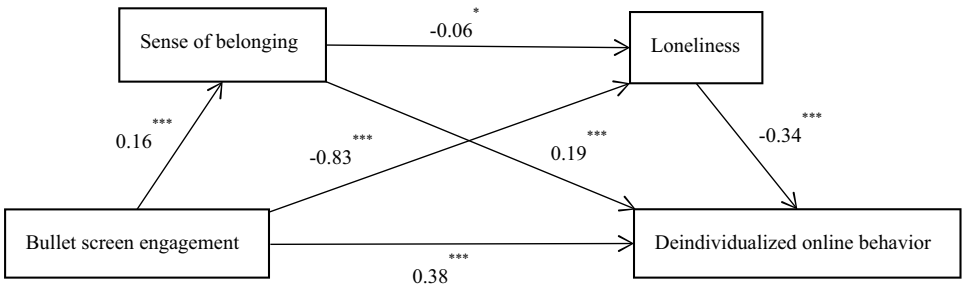

Fig. 2 Chain mediating role of belonging and loneliness between bullet screen engagement and deindividualized online behavior

mediating roles of increasing the sense of belonging and decreasing the sense of loneliness. Furthermore, both enhanced sense of belonging and decreased sense of loneliness have a chain-mediated effect on deindividualized online behavior.

Firstly, the study found that the degree of bullet-screen engagement positively predicts individuals' depersonalized online behaviors, which validates Hypothesis 1 and is consistent with the results of previous studies. For example, a study found that engagement and immersion in bullet screen in live video streaming can motivate individuals to engage in depersonalized behaviors such as swiping more gifts (Zhou et al., 2019). The results suggest that bullet-screens have the same potential for depersonalization as other communication platforms (Yi et al., 2019), and that malignant depersonalization triggered by instantaneous replies to bullet-screens may be more severe than the "building-style", time-delayed replies on microblogs, forums, and other platforms. Bullet screen has rich and interesting content and strong emotional rendering power. Such emotional involvement can promote their identification with virtual bullet-screen groups, thus promoting the generation of depersonalized online behaviors, such as publishing negative bullet screen information (Yan et al., 2023). However, the results of deindividualized online behaviors are not always contrary to general social norms (Lan & Zuo, 2009). If we can control the group norm in the bullet screen environment, set up the comment rule of the nature of the bottom line, and apply the bullet screen to the situation that needs the individual to produce the social identity, the participation and immersion of the bullet screen can play its dominant role, for example, setting the theme of bullet-screen by video type, such as the function of bullet-screen in teaching video, the involvement of bullet-screen content may improve the learning satisfaction and sense of learning situation involvement, and promote the consistent recognition of teaching content.

Second, the findings also found that sense of belonging can mediate the relationship between the level of bullet-screen engagement and deindividualized online behaviors, and the results verified Hypothesis 2. Engagement and immersion in bullet-screens do increase individuals' sense of belonging to the bullet-screen group (Xi et al., 2021), i.e., feeling identified and accepted by the bullet screen group, which leads to adherence to the default norms of the bullet screen group and occurrence of deindividualized online behaviors. For example, generating attitudes towards video content that are consistent with bullet-screen evaluations, and then liking the content to give it high ratings or retweeting and other behaviors, which is consistent with the results of existing research (Chen et al., 2019b). However, some other findings found that the higher the sense of belonging, the lower the probability of occurrence of deindividualized online behaviors, i.e., sense of belonging may be a negative predictor of deindividualized online behaviors (Chen et al., 2015). This is inconsistent with the results of this study. The reason may be the fact that

measures of deindividualized online behaviors in existing studies were not specific to the bullet-screen environment. Besides, the relationship between bullet-screen engagement and deindividualized online behavior will be regulated by the satisfaction degree of belonging of individuals in real life (Allen et al., 2021). In other words, if an individual's need for belonging can be met to a large extent in the real world, then the sense of belonging generated by the bullet-screen in video will not promote the generation of deindividualized online behavior, that is, the two are negatively correlated (Xiang & Chae, 2022). However, if an individual's need for belonging cannot be met in the real world, the bullet-screen in video will more easily promote their sense of belonging to the bullet-screen group, and this network form of belonging is directly related to the deindividualized online behavior.

In addition, the results of the study also found that reduced loneliness also has a mediating role between the level of bullet-screen engagement and deindividualized online behavior, and the results support Hypothesis 3, which is also consistent with the results of previous studies (He, 2017; Chen et al., 2019a). According to computer-mediated communication theory, the virtual situation created by bullet-screens can increase the sense of presence when individuals watch videos (Johnson, 2013), so the level of loneliness experienced by individuals under conditions of high engagement with bullet-screens is reduced. Previous studies have proven that social media such as bullet screens have greatly reduced people's loneliness and maintained their mental health during the COVID-19 pandemic (Gedvilaitė-Kordušienė & Mikulionienė, 2023). Combined with the results of the sense of belonging study, a low level of loneliness may already represent that individuals are better able to integrate into the bullet-screen environment and feel a sense of belonging to the bullet-screen group, and so develop a social identity, which leads to depersonalized online behaviors (Hayes et al., 2022).

Finally, sense of belonging and loneliness were found to have a chain mediating effect between bullet screen engagement level and deindividualized online behaviors, supporting Hypothesis 4. Increased sense of belonging negatively predicted decreased loneliness, consistent with the results of existing studies (Matthews et al., 2019; Theeke et al., 2015). According to the social identity theory of depersonalization effect, sense of belonging is important in the process of transforming personal identity into social identity (Greenaway et al., 2016), and it is an important influence mechanism for bullet-screens to reduce viewers' loneliness, when individuals begin to identify with a group, they will develop positive attitudes towards the group, which helps to reduce the individual's loneliness in online social interaction (Chen et al., 2019a). "Every man is not an island", especially when there are consistent views and attitudes in the network that are missing in reality, this sense of identification will greatly reduce the sense of loneliness and provide a prerequisite for the generation of depersonalized online behavior.

Considering the potential relationship between bullet-screen engagement and deindividualized online behavior, and the risk of deindividualized online behavior, this study discusses the relationship and mechanism between the two. The results of this study not only indicate that watching bullet screen videos may induce deindividualized behaviors, but also reveal part of the mechanism by which bullet screen engagement predicts deindividualized behaviors. The results expand the social identity theory of depersonalization effects, which can help individuals understand the function of bullet screen, avoid over-immersion, and depersonalize their words and behaviors. At the same time, the results also have some implications for developers to recognize and optimize the function of bullet-screens in video platforms.

The research also has limitations. Firstly, this study is a cross-sectional survey, which limits the ability to reveal causal relationships between bullet screen engagement and deindividualized online behavior compared to laboratory research and longitudinal tracking designs. In the future, the influence of different content and types of bullet screen on deindividualized online behavior in the real bullet screen environment can be studied. Secondly, the social identity theory of deindividuation effect suggests that deindividuation does not always lead to negative behavior. Future research could investigate the correlation between bullet screen engagement and specific deindividualized online behaviors, such as cyberbullying in the bullet screen environment or positive deindividuation online behaviors. For instance, some studies can examine the correlation between bullet screen engagement and online public welfare behavior, or investigate the direct relationship between bullet screen engagement and bullet screen group identity.

Conclusion

(1) Bullet screen engagement significantly and positively predicted deindividualized online behavior.
(2) Sense of belonging and loneliness mediated the association between bullet screen engagement and deindividualized online behavior.
(3) The relationship between bullet-screen engagement and deindividualized online behavior has a chain mediating effect of sense of belonging-loneliness.

Author contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Zou Mingxuan. The first draft of the manuscript was written by Mu Defang and Zou Mingxuan. All authors commented on previous versions of the manuscript and all authors read and approved the final manuscript.

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Data availability The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

The authors declare that they have no relevant or material financial interests that relate to the research described in this paper.

Disclosures All procedures were in accordance with the ethical standards of the responsible committee on human experimentation. Informed consent was obtained from all the participants for being included in the study.

Conflict of interest The authors declare that they have no conflict of interest.

References

Allen, K. A., Kern, M. L., Rozek, C. S., et al. (2021). Belonging: A review of conceptual issues, an integrative framework, and directions for future research. *Australian Journal of Psychology, 73*(1), 87–102.

Chen, J. (2019). *Research on the influencing factors of network cluster behavior under the mediating effect of emotion infection*. Yanshan University.

Chen, Y., Gao, Q., & Rau, P. L. P. (2015). Understanding gratifications of watching danmaku videos-videos with overlaid comments. *International Conference on Cross-Cultural Design* (pp. 153–163).

Chen, G., Zhou, S., & Zhi, T. (2019a). Viewing mechanism of lonely audience: Evidence from an eye movement experiment on barrage video. *Computers in Human Behavior, 101*, 327–333.

Chen, Y. X., Wang, S. J., & Liu, X. P. (2019b). Relatedness need-satisfaction and deindividuation internet behaviors in adolescents: A moderated mediating model. *Chinese Journal of Clinical Psychology, 27*(06), 1256–1259.

Gedvilaitė-Kordušienė, M., & Mikulionienė, S. (2023). Perceived effects of the COVID-19 pandemic on loneliness: The most vulnerable population groups. *Emerging Science Journal, 7*, 281–293.

Gong, H., & Yan, D. (2023). The impact of Danmaku-based and synchronous peer feedback on L2 oral performance: A mixed-method investigation. *PLoS ONE, 18*(4), 1–28.

Greenaway, K. H., Cruwys, T., Haslam, S. A., et al. (2016). Social identities promote well-being because they satisfy global psychological needs. *European Journal of Social Psychology, 46*(3), 294–307.

Hagerty, B., & Patusky, K. (1995). Developing a measure of sense of belonging. *Nursing Research, 44*(1), 9–13.

Hao, X., Xu, S., & Zhang, X. (2019). Barrage participation and feedback in travel reality shows: The effects of media on destination image among generation y. *Journal of Destination Marketing & Management, 12*, 27–36.

Hawkley, L. C., & Cacioppo, J. T. (2010). Loneliness matters: A theoretical and empirical review of consequences and mechanisms. *Annals of Behavioral Medicine, 40*(2), 218–227.

Hayes, A. F. (2017). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach*. Guilford.

Hayes, S., Carlyle, M., Haslam, S. A., et al. (2022). Exploring links between social identity, emotion regulation, and loneliness in those with and without a history of mental illness. *British Journal of Clinical Psychology, 61*(3), 701–734.

He, Y. L. (2017). *The relationship of college students' bullet-screen using motivation, immersion and loneliness*. Huazhong University of Science and Technology.

Johnson, D. (2013). Polyphonic/pseudo-synchronic: Animated writing in the comment feed of nicovideo. *Japanese Studies, 33*(3), 297–313.

Lan, Y. J., & Zuo, B. (2009). An introduction to social identity model of deindividuation effects. *Advances in Psychological Science, 17*(02), 467–472.

Li, R. S. (2020). *Research on the impact of Danmaku on learners with different cognitive styles in online video teaching* (Unpublished master's thesis). Southwest University

Lim, M. H., Allen, K. A., Furlong, M. J., et al. (2021). Introducing a dual continuum model of belonging and loneliness. *Australian Journal of Psychology, 73*(1), 81–86.

Matthews, T., Danese, A., Caspi, A., et al. (2019). Lonely young adults in modern Britain: Findings from an epidemiological cohort study. *Psychological Medicine, 49*(2), 268–277.

Planalp, S., & Rosenberg, J. (2014). Emotion in interpersonal communication. In C. R. Bergereds (Ed.), *Interpersonal Communication* (vol 6, pp. 273–296). Berlin, Boston: De Gruyter Mouton.

Shih, H. P., & Huang, E. (2014). Influences of web interactivity and social identity and bonds on the quality of online discussion in a virtual community. *Information Systems Frontiers, 16*(4), 627–641.

Spears, R. (2017). Deindividuation. In S. G. Harkins, K. D. Williams, & J. M. Burger (Eds.), *The Oxford handbook of social influence* (pp. 279–297). Oxford University Press.

Theeke, L. A., Mallow, J., Gianni, C., et al. (2015). The experience of older women living with loneliness and chronic conditions in Appalachia. *Journal of Rural Mental Health, 39*(2), 61–72.

Vallury, K. D., Baird, B., Miller, E., & Ward, P. (2021). Going viral: Researching safely on social media. *Journal of Medical Internet Research, 23*(12), e29737.

Xi, D., Xu, W., Chen, R., et al. (2021). Sending or not? A multimodal framework for danmaku comment prediction. *Information Processing & Management, 58*(6), 102687.

Xiang, Y., & Chae, S. W. (2022). Influence of perceived interactivity on continuous use intentions on the danmaku video sharing platform: Belongingness perspective. *International Journal of Human–Computer Interaction, 38*(6), 573–593.

Yan, Z., Yang, Z., & Griffiths, M. D. (2023). 《Danmu》 preference, problematic online video watching, loneliness and personality: An eye-tracking study and survey study. *BMC Psychiatry, 23*(1), 523.

Yang, Y. (2020). The danmaku interface on bilibili and the recontextualised translation practice: A semiotic technology perspective. *Social Semiotics, 30*(2), 254–273.

Yawei, B. (2021). New features of youth patriotic expression in the new media environment: Taking the bullet screen culture of station B as an example. *China Youth Study, 07*, 96–101.

Yi, X. M., Zhang, A. J., & Yao, Y. T. (2019). Rationality and hubbub: A survey of college students' involvement in micro-blog violence. *China Youth Studies, 03*, 102–107. 73.

Zeng, Q., Guo, Q., Zhuang, W., Zhang, Y., & Fan, W. (2023). Do real-time reviews matter? Examining how bullet screen influences consumers' purchase intention in live streaming commerce. *Information Systems Frontiers, 25*(5), 2051–2067.

Zhou, H., & Long, L. R. (2004). Statistical remedies for common method biases. *Advances in Psychological Science, 06*, 942–950.

Zhou, J., Zhou, J., Ding, Y., et al. (2019). The magic of danmaku: A social interaction perspective of gift sending on live streaming platforms. *Electronic Commerce Research and Applications, 34*, 100815.

Zimbardo, P. G. (1969). The human choice: Individuation, reason, and order versus deindividuation, impulse, and chaos. *Nebraska Symposium on Motivation, 17*, 237–307.

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